

Topic 22 Personalization and Recommendation Under Privacy Constraints

Project structure, datasets, metrics, literature alignment, and deep research findings for Model 2

Overall conclusion. This topic is feasible and research-worthy. The course requirement should be handled explicitly: build four models and reproduce one published baseline separately. Use SASRec as the common sequential reference, but do not count it as both the reproduced baseline and Model 1. The final comparison should study centralized, federated, on-device, and differentially private systems under one controlled evaluation protocol.

Recommended final model structure

Role	Recommended model	Main paper and alignment
Separately reproduced baseline	SASRec	Self-Attentive Sequential Recommendation . Predicts the next item from a user's chronological history using self-attention. This becomes the common quality and architecture reference.
Model 1: Centralized sequential	BERT4Rec	BERT4Rec . Uses bidirectional Transformer attention and masked-item prediction. It represents the centralized quality ceiling and is directly comparable with SASRec on common sequential datasets.
Model 2: Federated	FedSASRec with secure aggregation	Use the federated sequential strategy from CF-FedSR together with the secure aggregation protocol from Bonawitz et al. User data remains local; only securely aggregated model updates are shared.
Model 3: On-device	Compressed/on-device SASRec	On-Device Next-Item Recommendation with Self-Supervised Knowledge Distillation . Compresses the recommendation model and uses self-supervised knowledge distillation to preserve capacity on edge devices.
Model 4: Differentially private	DP-SASRec using user-level DP-SGD	Use DIPSGNN as the main sequential differential-privacy reference and DP-SGD for implementation. Add gradient clipping and calibrated Gaussian noise while reporting (epsilon, delta).

BPR-MF may be included as an additional classical comparison, but it should not be the main baseline because it ignores interaction order. The project should preserve one common SASRec reference so the privacy and systems trade-offs remain interpretable.

1. Paper alignment and implementation scope

The papers do not all use exactly the same architecture. CF-FedSR uses a GRU-based sequential recommender, although its federated framework is model-agnostic. DIPSGNN uses a graph neural network rather than SASRec. The on-device paper is the closest match for the compressed/on-device direction. This creates an important methodological distinction.

Paper reproduction. Reproduce the published method as described, including its architecture and reported assumptions where practical. Fair common-backbone experiment. Implement federated SASRec and DP-SASRec so every privacy-constrained approach can also be compared against the same SASRec baseline. This is scientifically stronger than claiming that all papers use identical models.

What each model delivers

Model	Intuitive functionality	Comparison with SASRec
SASRec baseline	A central server collects all user histories and predicts the next item.	Establishes the normal accuracy ceiling without privacy restrictions.
BERT4Rec	Looks at both sides of masked historical sequences to learn richer user preferences.	Measures whether a stronger centralized model improves accuracy before privacy is introduced.
FedSASRec	Each simulated device trains on its own user history. The server receives only aggregated updates.	Measures the accuracy and communication cost of keeping raw data decentralized.
On-device SASRec	A compressed model is downloaded to a device and performs local inference or small local updates.	Measures the trade-off between accuracy, model size, latency, memory, and communication.
DP-SASRec	Adds mathematically calibrated noise to limit how much any individual user affects the learned model.	Measures the accuracy loss as the privacy budget becomes stronger.

Federated learning alone is not a complete privacy guarantee. Gradients or model updates can leak information, so the secure-aggregation layer and the privacy analysis from FedMF are important for the threat model and failure analysis. Secure aggregation protects the visibility of individual updates to the server; it does not automatically provide a formal epsilon/delta guarantee.

2. Datasets and recommended scope

Dataset	Why it is appropriate	Dataset link
MovieLens-1M	Best primary dataset: manageable size, timestamps, user histories, and broad use in sequential recommendation research.	GroupLens MovieLens-1M
Amazon Beauty 5-core	Realistic, sparse e-commerce interactions. Useful for long-tail behavior, sparse histories, and cold-start analysis.	UCSD Amazon datasets
RetailRocket	Useful for reproducing a session-based or on-device recommendation setting if time and preprocessing capacity permit.	RetailRocket dataset
Tmall	Large-scale e-commerce sequence data used as an optional stress test. It should not be required for the core project.	Tmall dataset

Recommended scope: Use MovieLens-1M as the common benchmark for all four models and Amazon Beauty as the secondary robustness dataset. Use Tmall or RetailRocket only if time and hardware permit. This keeps the core comparison reproducible while still testing one sparse e-commerce setting.

3. Metrics to report

Category	Metrics to report	Why it matters
Recommendation quality	Recall@5, Recall@10, NDCG@5, NDCG@10, MRR@10, Hit Rate@K	Shows top-K ranking quality and makes the accuracy gap between centralized and privacy-constrained models measurable.
Privacy and leakage	Differential privacy budget (epsilon, delta); membership-inference attack success or AUC; whether raw data reaches the server; whether individual gradients are visible.	Separates data locality from formal privacy. Federated learning and secure aggregation should not be reported as equivalent to DP.
System efficiency	Communication bytes per round; total bytes; number of communication rounds; training time; model size in MB; inference latency; memory usage; local-device update time.	Captures the practical costs of federation, compression, and on-device execution.
Robustness and fairness	Cold-start user performance; long-tail item Recall/NDCG; average and worst-user performance; standard deviation across users; performance across multiple random seeds.	Prevents a higher average score from hiding harm to sparse, cold-start, or low-data users.

The existing Model 2 metrics are retained here: Recall@5/10, NDCG@5/10, MRR@10, Hit Rate@K, epsilon/delta, membership-inference success or AUC, raw-data exposure, individual-gradient visibility, communication bytes, rounds, training time, model size, latency, memory, local update time, cold-start performance, long-tail performance, worst-user performance, standard deviation, and multi-seed error bars.

4. Fair experimental design

Keep the following controls identical across all models so that differences can be attributed to architecture, federation, compression, or privacy rather than preprocessing or evaluation choices.

Control	Requirement across all models
Data and vocabulary	Use the same dataset, item vocabulary, filtering rules, and interaction semantics.
Temporal protocol	Use the same chronological train/validation/test split and never mix future interactions into training.
Model inputs	Use the same maximum sequence length and the same embedding dimension where possible; document unavoidable differences.
Evaluation	Use identical negative-sampling or candidate-ranking procedures and identical evaluation code.
Statistics	Run at least five random seeds and report confidence intervals or error bars.

Chronological split

Earlier interactions are used for training, the second-to-last interaction for validation, and the last interaction for testing. Do not randomly mix future interactions into training because that causes temporal leakage. For Model 2, each user can be simulated as one client while keeping the test population fixed across all variants.

Two-semester feasibility

Semester	Deliverables
298A	Literature review; data pipeline and fixed benchmark; reproduce SASRec; build centralized BERT4Rec; establish evaluation, logging, and reproducibility infrastructure.
298B	Implement FedSASRec with secure aggregation; implement compressed/on-device SASRec; implement DP-SASRec; run membership-inference experiments; compare accuracy, privacy, communication, latency, and cold-start behavior; build the final web interface and failure-analysis report.

This is sufficiently complex for the course because it combines sequential recommendation, federated learning, secure aggregation, model compression, differential privacy, privacy attacks, and system benchmarking. The project remains feasible if MovieLens-1M and Amazon Beauty are treated as the core scope and larger stress-test datasets remain optional.

5. Deep research findings for Model 2

Model 2 should be implemented as FedSASRec with secure aggregation, while the base model remains centralized SASRec. Centralized SASRec trains on all user histories in one server, so it should provide the strongest and fastest reference performance because it has complete access to the data and gradients. FedSASRec keeps each user's sequence on their device and trains locally, sending only masked updates whose aggregate is used to update the global model. This improves data locality and can support local personalization, but it may reduce accuracy or slow convergence because user data is highly non-IID, sequences are sparse, clients may drop out, and communication is expensive. Secure aggregation hides each individual client update from the server, but it is not differential privacy and does not by itself protect against every inference or collusion attack.

The strongest directly relevant method is CF-FedSR, which adds adaptive client selection, user clustering, fairness-aware aggregation, and local personalization. It reported approximately 8.90% higher HR@10, 15.46% higher NDCG@10, and 10.67% fewer communication rounds than FedAvg on Amazon Beauty. However, it uses a GRU4Rec backbone rather than SASRec, so these numbers should not be claimed as a direct FedSASRec-versus-SASRec result. The fair comparison should therefore use centralized SASRec, FedAvg-SASRec, FedSASRec with secure aggregation, and then CF-FedSR-inspired SASRec variants. The research finding should be whether federated training approaches centralized SASRec while improving data locality, and whether clustering or personalization recovers accuracy and communication losses.

Dimension	Base: centralized SASRec	Model 2: FedSASRec with secure aggregation
Training location	All histories and gradients are available to one central trainer.	Each user is a client. Training examples remain on that client; the server broadcasts a global model and receives updates.
Ranking quality	Reference quality ceiling for the project because the server can learn from all users jointly.	May lose Recall@K/NDCG@K or converge more slowly under non-IID data, sparse sequences, client dropout, and local training.
Privacy exposure	Raw user sequences are collected and processed centrally.	Raw sequences remain local. Secure aggregation is intended to reveal only an aggregate update, not an individual update.
Formal guarantee	No data-locality guarantee; the server sees the underlying training data.	Secure aggregation is not differential privacy. It does not by itself protect against all collusion, malicious-client, or aggregate-leakage attacks.
Personalization	One global model learns from all users but has no local-only adaptation by default.	Local fine-tuning, local adapters, or cluster-level models can address heterogeneous user interests.
Communication	No federated client-server rounds; centralized data transfer is assumed.	Repeated model download and update upload. Secure aggregation adds protocol overhead; measure bytes and rounds explicitly.
Compute	Training is concentrated on server hardware.	Training is distributed across simulated clients or devices; local epochs affect convergence and device cost.
Fairness	Aggregate ranking metrics can hide weak performance for sparse users.	Report mean, worst-decile, variance, or IQR across clients; CF-FedSR explicitly targets fairness.

Evidence and limitations from the available papers

CF-FedSR is the best algorithmic reference for the project because it treats communication, fairness, and personalization as first-class federated problems. Bonawitz et al. provide the secure-aggregation mechanism needed to keep individual updates hidden from the server. Chai et al. show why raw-data locality alone is insufficient: gradients may still leak raw preferences. The recent FedSR survey places FedSASRec and related systems in a broader communication and personalization literature; it also reports that parameter-transmission approaches can be substantially more expensive than parameter-free or compressed alternatives. These sources support the design, but the final numerical claims must come from the common SASRec implementation and controlled seeds.

6. Final recommendation

Project title: Privacy-Utility Trade-offs in Sequential Recommendation: A Comparative Study of Centralized, Federated, On-Device, and Differentially Private SASRec-Based Systems.

Use SASRec as the separately reproduced baseline, BERT4Rec as Model 1, FedSASRec with secure aggregation as Model 2, compressed/on-device SASRec as Model 3, and DP-SASRec with user-level DP-SGD as Model 4. Keep paper-specific reproductions separate from the common-backbone comparison and report accuracy, privacy, communication, latency, fairness, and cold-start behavior together.

Research paper links

Source	Use in the project
SASRec	Separately reproduced sequential baseline and common architecture reference.
BERT4Rec	Centralized Transformer comparison using masked-item prediction.
CF-FedSR	Closest federated sequential reference for client selection, clustering, fairness, and personalization. Uses GRU4Rec, so adapt carefully.
Practical Secure Aggregation	Protocol reference for hiding individual client updates from the server while aggregating them.
Secure Federated Matrix Factorization	Privacy warning: gradients can leak raw preferences; federated learning alone is not a complete privacy guarantee.
On-Device Next-Item Recommendation	Compression and self-supervised knowledge distillation reference for Model 3.
DIPSGNN	Sequential differential-privacy reference; uses a noisy GNN rather than SASRec.
DP-SGD	Implementation foundation for clipping, Gaussian noise, and privacy-accounting experiments.
PTF-FSR	Recent communication-efficient federated sequential direction; useful as related work, not required for the core build.
FedSR survey	Field map and reported communication/fairness comparisons across federated sequential systems.

The strongest defensible contribution is not claiming a new federated architecture from scratch. It is a controlled, reproducible comparison showing how much sequential recommendation quality is lost or retained when the same SASRec reference is trained under different privacy and deployment constraints, with explicit measurement of leakage, communication, fairness, and failure modes.